

Lab 2:

Perform the following DB operations using Cassandra.

- a) Create a keyspace by name Employee
- b) Create a column family Employee_Info with attributes: Emp_Id (PK), Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name
- c) Insert values into the table in batch
- d) Update Employee name and Department of EmpId 121
- e) Sort Employee records based on salary
- f) Alter the schema to add a column Projects (SET)
- g) Update the altered table to add project names
- h) Create a TTL of 15 seconds to display values of Employees

```
CREATE KEYSPACE IF NOT EXISTS Employee
  WITH replication={'class': 'SimpleStrategy', 'replication_factor': 1};
USE Employee;
```

```
CREATE TABLE IF NOT EXISTS Employee_Info (
  Emp_Id          INT PRIMARY KEY,
  Emp_Name        TEXT,
  Designation     TEXT,
  Date_of_Joining DATE,
  Salary          DOUBLE,
  Dept_Name       TEXT
);
```

```
BEGIN BATCH
  INSERT INTO Employee_Info(Emp_Id,Emp_Name,Designation,Date_of_Joining,Salary,Dept_Name)
    VALUES(121,'Tarun S','Manager','2026-02-06',90000,'HR');
  INSERT INTO Employee_Info(Emp_Id,Emp_Name,Designation,Date_of_Joining,Salary,Dept_Name)
    VALUES(122,'Tanu R','Developer','2026-02-06',75000,'IT');
  INSERT INTO Employee_Info(Emp_Id,Emp_Name,Designation,Date_of_Joining,Salary,Dept_Name)
    VALUES(123,'Tejas K','Analyst','2026-02-06',65000,'IT');
APPLY BATCH;
```

```
UPDATE Employee_Info
  SET Emp_Name='TR Sunadoli', Dept_Name='Finance'
  WHERE Emp_Id=121;
```

```
CREATE TABLE IF NOT EXISTS Employee_By_Dept (
  Dept_Name TEXT, Salary DOUBLE, Emp_Id INT,
  Emp_Name TEXT, Designation TEXT, Date_of_Joining DATE,
  PRIMARY KEY(Dept_Name,Salary,Emp_Id)
) WITH CLUSTERING ORDER BY(Salary DESC, Emp_Id ASC);
```

```
BEGIN BATCH
  INSERT INTO Employee_By_Dept VALUES('HR',90000,121,'TR Sunadoli','Manager','2026-02-06');
  INSERT INTO Employee_By_Dept VALUES('IT',75000,122,'Tanu R','Developer','2026-02-06');
  INSERT INTO Employee_By_Dept VALUES('IT',65000,123,'Tejas K','Analyst','2026-02-06');
APPLY BATCH;

SELECT * FROM Employee_By_Dept WHERE Dept_Name='IT';
```

```
ALTER TABLE Employee_Info ADD Projects SET<TEXT>;
```

```
UPDATE Employee_Info SET Projects={'ERP System','HR Portal'} WHERE Emp_Id=121;
```

```
INSERT INTO Employee_Info(Emp_Id,Emp_Name,Designation,Date_of_Joining,Salary,Dept_Name)
VALUES(124,'Vasishta M','Tester','2026-02-06',55000,'QA') USING TTL 15;
```

```
SELECT * FROM Employee_Info;
```

emp_id	date_of_joining	dept_name	designation	emp_name	salary
123	2026-02-06	IT	Analyst	Tejas K	65000
122	2026-02-06	IT	Developer	Tanu R	75000
121	2026-02-06	Finance	Manager	TR Sunadoli	90000

(3 rows)

```
// After alter and update:
```

emp_id	date_of_joining	dept_name	emp_name	projects	salary
123	2026-02-06	IT	Tejas K		65000
122	2026-02-06	IT	Tanu R		75000
121	2026-02-06	Finance	TR Sunadoli	{'ERP System','HR Portal'}	90000

(3 rows)

```
// Sort by salary (IT dept):
```

dept_name	salary	emp_id	date_of_joining	designation	emp_name
IT	75000	122	2026-02-06	Developer	Tanu R
IT	65000	123	2026-02-06	Analyst	Tejas K

(2 rows)